Notes: Gromov's Almost Flat Manifold Theorem and Its Generalizations Based on lectures of Jikang Wang in USTC

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Contents

Notations and main theorems **2**

1 Equivariant Gromov–Hausdorff convergence and local groups **3**
1.1 Gromov–Hausdorff approximations . . . 3
1.2 Equivariant approximations . . . 3
1.3 Precompactness . . . 4
1.4 Local groups . . . 4
1.5 Constructing a group from a local group . . . 4
1.6 The local-determines-global property . . . 5
1.7 Subgroups converging to the identity component . . . 5
1.8 Commutator estimates . . . 5

2 The Bieberbach theorem **6**
2.1 Set-up . . . 6
2.2 Controlling the nilpotency steps . . . 6
2.3 No rotation . . . 7

3 Gromov's theorem **7**
3.1 Set-up: pseudo covers and local groups . . . 7
3.2 Commutator estimates . . . 8
3.3 Controlling the nilpotency steps . . . 9
3.4 Small rotations . . . 9
3.5 Constructing a nilpotent basis . . . 10
3.6 Constructing a nilmanifold . . . 11

4 Ricci curvature **11**
4.1 Set-up . . . 11
4.2 Escape norm . . . 11

Recall

semi-direct product of two gps:$N \ltimes H$
Assume $N \triangleleft H$, define $N \ltimes H$ by setting the product to be $(n_1, h_1) * (n_2, h_2) = (n_1 * \phi_{h_1}(n_2), h_1 h_2)$
ϕ_{h_i} is the element h_i act on element of N, $H \curvearrowright N$ in some way. $\phi : H \longrightarrow Aut(N)$;
Killing field: if vector field X satisfying $L_X g = 0$, then X is called killing field.

Notations and main theorems

M is a compact n-dimensional Riemannian manifold. We use $\varepsilon > 0$ for a small number, C for a constant, and $\varepsilon_i \to 0$ as $i \to \infty$.

Nilpotent groups. Let G be a group and $H < G$. Write

$$[G, H] = \langle [g, h] : g \in G,\ h \in H \rangle, \qquad [g, h] = ghg^{-1}h^{-1}.$$

Let

$$G_1 = G, \qquad G_2 = [G, G], \qquad G_3 = [G, G_2], \qquad \ldots.$$

G is nilpotent if $G_k = \{e\}$ for some k.

Theorem 1 (First Bieberbach theorem). If M is compact and $\mathrm{Sec}_M = 0$, then a finite covering space of M is isometric to a torus $\mathbb{R}^n/\mathbb{Z}^n$.

Theorem 2 (Gromov). There exists $\varepsilon(n) > 0$ such that, if

$$\mathrm{diam}(M) \leq \varepsilon(n), \qquad |\,\mathrm{Sec}_M\,| \leq \varepsilon(n),$$

then a finite covering space of M is diffeomorphic to a nilmanifold N/Γ. Here N is a connected, simply connected nilpotent Lie group, and Γ is a lattice.

Remark. N is diffeomorphic to $\mathbb{R}^n$. A lattice Γ is a discrete subgroup of N such that N/Γ is compact.

Theorem 3 (Recent work). Fix $v > 0$. There exists $\varepsilon(n, v) > 0$ such that, if

$$\mathrm{diam}(M) < \varepsilon(n, v), \qquad \mathrm{Ric}_M \geq -(n-1), \qquad \mathrm{Vol}(B_1(\tilde{p})) \geq v,$$

where $\tilde{p}$ is a point in the universal cover of M, then a finite cover of M is diffeomorphic to a nilmanifold.

Remarks.

1. In each theorem, the order of the finite covering can be bounded by a constant $C(n)$.
2. Finite covering spaces correspond to the rotational part. In Theorem 1,

   $$\pi_1(M, p) < O(n) \ltimes \mathbb{R}^n, \qquad \Gamma = \pi_1(M, p) \cap \mathbb{R}^n.$$

   In Theorems 2 and 3,

   $$\pi_1(M, p) < \mathrm{Aut}(N) \ltimes N, \qquad \Gamma = \pi_1(M, p) \cap N.$$

3. In Theorem 2, $\widetilde{M}$ is non-collapsing.
4. M is almost flat if

   $$\mathrm{diam}(M)^2 \max |\,\mathrm{Sec}_M\,|$$

   is small. This quantity is invariant under scaling the metric.

1 Equivariant Gromov–Hausdorff convergence and local groups

1.1 Gromov–Hausdorff approximations

Definition 1.1. Let X, Y be compact metric spaces. We say that X is ε-close to Y if there is a map $f : X \to Y$ such that

1. for all $x, x' \in X$,
$$\big|d(f(x), f(x')) - d(x, x')\big| < \varepsilon;$$
2. $f(X)$ is ε-dense in Y.

Such an f is called an ε-Gromov–Hausdorff approximation (ε-GHA). We write $X_i \xrightarrow{\text{GH}} X$ if X_i is ε_i-close to X with $\varepsilon_i \to 0$.

Definition 1.2. For pointed metric spaces (X, p) and (Y, q), we say that (X, p) is ε-close to (Y, q) if there is an ε-GHA
$$f : B_{1/\varepsilon}(p) \longrightarrow B_{1/\varepsilon}(q).$$

1.2 Equivariant approximations

Definition 1.3. Assume $G < \text{Isom}(X)$ and $H < \text{Isom}(Y)$. Let
$$G(p, 1/\varepsilon) = \{g \in G : d(p, gp) \le 1/\varepsilon\}.$$
We say that (X, p, G) is ε-eGHA close to (Y, q, H) if there are maps (f, φ, ψ) such that

1. $f : B_{1/\varepsilon}(p) \to B_{1/\varepsilon}(q)$ is an ε-GHA;
2. $\varphi : G(p, 1/\varepsilon) \to H(q, 1/\varepsilon)$ satisfies
$$d\big(f(gx), \varphi(g)f(x)\big) < \varepsilon;$$
3. $\psi : H(q, 1/\varepsilon) \to G(p, 1/\varepsilon)$ satisfies
$$d\big(hf(x), f(\psi(h)x)\big) < \varepsilon.$$

We write
$$(X_i, p_i, G_i) \xrightarrow{\text{eGH}} (X, p, G)$$
if there are ε_i-eGHAs with $\varepsilon_i \to 0$.

Remark 1.4. φ and ψ are almost homomorphisms: $\varphi(g_1)\varphi(g_2)$ is ε-close to $\varphi(g_1 g_2)$.

Example 1.5. Let
$$M_i = iS^1 \times iS^1, \qquad G_i = \left\langle \left(r_{2\pi/i^2}, r_{2\pi/i^3} \right) \right\rangle \cong \mathbb{Z}_{i^3},$$
where iS^1 is the circle of radius i and r_θ is a rotation. Then
$$(M_i, p_i, G_i) \xrightarrow{\text{eGH}} (\mathbb{R}^2, 0, \mathbb{R} \times \mathbb{Z}).$$

1.3 Precompactness

Theorem 1.6 (Gromov)**.** *Assume that (M_i, p_i) are complete n-dimensional manifolds with* $\mathrm{Ric}_{M_i} \geq -(n-1)$*. After passing to a subsequence,*

$$(M_i, p_i) \xrightarrow{\mathrm{GH}} (X, p).$$

Remark 1.7. X is called a Ricci limit space. It is non-collapsing if there is $v > 0$ such that $\mathrm{Vol}(B_1(p_i)) \geq v$; otherwise it is collapsing.

Theorem 1.8 (Equivariant precompactness, Fukaya–Yamaguchi)**.** *Assume $(X_i, p_i) \xrightarrow{\mathrm{GH}} (X, p)$ and $G_i < \mathrm{Isom}(X_i)$. After passing to a subsequence, there is $G < \mathrm{Isom}(X)$ such that*

$$\begin{array}{ccc} (X_i, p_i, G_i) & \xrightarrow{\mathrm{eGH}} & (X, p, G) \\ {\scriptstyle \pi_i}\big\downarrow & & \big\downarrow{\scriptstyle \pi} \\ (X_i/G_i, \bar{p}_i) & \xrightarrow{\mathrm{GH}} & (X/G, \bar{p}). \end{array}$$

Theorem 1.9 (Cheeger–Colding; Colding–Naber)**.** *If X is a Ricci limit space, then* $\mathrm{Isom}(X)$ *is a Lie group.*

Theorem 1.10 (Volume convergence, Colding)**.** *If X is a non-collapsing Ricci limit space, then*

$$\mathrm{Vol}(B_1(p_i)) \longrightarrow \mathrm{Haus}^n(B_1(p)).$$

1.4 Local groups

Definition 1.11. A local group G is a topological space with $e \in G$, an inverse map $G \to G$, and a partially defined product $\Omega \to G$, where

1. Ω is an open neighborhood of $(G \times \{e\}) \cup (\{e\} \times G)$ in $G \times G$;
2. the inverse and product maps are continuous;
3. if $(gh)k$ and $g(hk)$ are defined, then $(gh)k = g(hk)$;
4. $gg^{-1} = e$, $ge = g$, and $eg = g$.

Example 1.12. If $G < \mathrm{Isom}(X)$, then

$$G(p, 1/\varepsilon) = \{g \in G : d(p, gp) \leq 1/\varepsilon\}$$

is a local group.

Definition 1.13. A map $\varphi : G \to H$ between local groups is a homomorphism if

$$\varphi(g_1)\varphi(g_2) = \varphi(g_1 g_2)$$

whenever the product is defined.

1.5 Constructing a group from a local group

Let F be the free group generated by e_g, $g \in G$. Set

$$\widehat{G} = F/\!\sim, \qquad e_{g_1} e_{g_2} \sim e_{g_1 g_2} \quad \text{if } g_1, g_2, g_1 g_2 \in G.$$

There is a natural homomorphism

$$\iota : G \longrightarrow \widehat{G}, \qquad \iota(g) = [g].$$

For a group H and a homomorphism $\varphi : G \to H$, there is a unique homomorphism $\widehat{\varphi} : \widehat{G} \to H$ such that $\varphi = \widehat{\varphi} \circ \iota$:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & H \\ {\scriptstyle\iota}\downarrow & & \| \\ \widehat{G} & \xrightarrow{\widehat{\varphi}} & H. \end{array}$$

Example 1.14. For the local group $G_i(p_i, 20)$ in the preceding torus example, $\widehat{G}_i = \mathbb{Z}^2$.

Lemma 1.15. *If G is a subset of a group H, then $\iota : G \to \widehat{G}$ is injective.*

Proof. Let $\varphi : G \to H$ be the inclusion map. Then

$$\ker \varphi = \{e\}, \qquad \varphi = \widehat{\varphi} \circ \iota \quad \Longrightarrow \quad \ker \iota = \{e\}.$$

□

1.6 The local-determines-global property

Definition 1.16. Let Γ be a group and let $S \subset \Gamma$ be symmetric and contain e. We say that S defines Γ if $\widehat{S}$ is naturally isomorphic to Γ.

Theorem 1.17. *Assume $\operatorname{diam}(M) \le D$, and let $(\widetilde{M}, \tilde{p}, \Gamma)$ be its universal cover with the deck transformation group. Then $\Gamma(\tilde{p}, 10D)$ defines Γ.*

1.7 Subgroups converging to the identity component

Theorem 1.18 (Fukaya–Yamaguchi)**.** *Assume $\operatorname{Ric}_{M_i} \ge -(n-1)$, $\operatorname{diam}(M_i) \le D$, and*

$$(\widetilde{M}_i, \tilde{p}_i, \Gamma_i) \xrightarrow{\text{eGH}} (X, p, G).$$

Let G_0 be the identity component of G. Then there are $\Gamma_i' \triangleleft \Gamma_i$ such that

$$\Gamma_i' \xrightarrow{\text{eGH}} G_0, \qquad \Gamma_i/\Gamma_i' \cong G/G_0.$$

Proof sketch. The theorem holds for local groups. The local-determines-global property gives the global statement. □

1.8 Commutator estimates

Lemma 1.19. *Let G be a Lie group with a left-invariant metric. There are a neighborhood U of e and $C > 0$ such that, for all $g, h \in U$,*

$$d(e, [g, h]) \le C\, d(e, g) d(e, h).$$

Proof sketch. Take a small neighborhood V of 0 in the Lie algebra such that $\exp : V \to U$ is bi-Lipschitz. For $u \in V$, $d(e, \exp u) \simeq \|u\|$. By BCH,

$$\log[\exp u, \exp v] = [u, v] + \text{higher-order terms}.$$

The bracket is bilinear and $\|[u, v]\| \le C\|u\|\|v\|$. □

Corollary 1.20. *There is a neighborhood U' of e such that every discrete subgroup $\Gamma < G$ satisfying $\langle \Gamma \cap U' \rangle = \Gamma$ is nilpotent.*

Proof sketch. Take $C > 1$ and $\delta < C^{-2}$ with $B_\delta(e) \subset U$, and set $U' = B_\delta(e)$. For $g_1, g_2, \ldots, g_k \in U'$,

$$[g_1, g_2] \in B_{\delta/C}(e), \qquad [[g_1, g_2], g_3] \in B_{\delta/C^2}(e), \qquad \ldots.$$

Since Γ is discrete, it is nilpotent. □

2 The Bieberbach theorem

2.1 Set-up

Let M be compact with $\mathrm{Sec}_M = 0$, and let $M_i = \frac{1}{i}M$. Then

$$\begin{array}{ccc} (\widetilde{M}_i, \tilde{p}_i, \Gamma_i) & \xrightarrow{\text{eGH}} & (\mathbb{R}^n, 0, G) \\ \downarrow & & \downarrow \\ M_i & \xrightarrow{\text{GH}} & \{\text{pt}\}, \end{array}$$

where $\Gamma_i = \pi_1(M_i, p_i)$, and there are $\Gamma_i' \triangleleft \Gamma_i$ with $\Gamma_i/\Gamma_i' \cong G/G_0$.

Lemma 2.1. *G/G_0 is finite.*

2.2 Controlling the nilpotency steps

We have

$$\widetilde{M}_i = \mathbb{R}^n, \qquad \Gamma_i' < O(n) \ltimes \mathbb{R}^n, \qquad \Gamma_i' = \langle \Gamma_i' \cap B_\varepsilon(e) \rangle.$$

Thus Γ_i' is nilpotent. Let

$$\pi_{O(n)} : O(n) \ltimes \mathbb{R}^n \to O(n), \qquad K_i = \pi_{O(n)}(\Gamma_i').$$

Then $K_i = \langle K_i \cap B_\varepsilon(e) \rangle$ is nilpotent.

Lemma 2.2. *Let $g \in O(n)$ and $h \in B_{0.1}(e) \subset O(n)$. If $[g,[g,h]] = e$, then $[g,h] = e$.*

Proof sketch. Decompose $\mathbb{C}^n = V_1 \oplus \cdots \oplus V_r$ into the eigenspaces of g. The relation $[g,[g,h]] = e$ implies that $h^{-1}V_j$ is g-invariant, hence

$$h^{-1}V_j = (h^{-1}V_j \cap V_1) \oplus \cdots \oplus (h^{-1}V_j \cap V_r).$$

Since $h \in B_{0.1}(e)$, $h^{-1}V_j = V_j$. On V_j, g is scalar, so $hg = gh$ on V_j. Therefore $hg = gh$. □

Corollary 2.3. *K_i is abelian.*

Proof sketch. For $g, h \in K_i \cap B_\varepsilon(e)$, nilpotency gives

$$[g,[g,\ldots,[g,h]\ldots]] = e.$$

Applying the lemma gives $gh = hg$. □

Corollary 2.4. $[[\Gamma_i', \Gamma_i'], \Gamma_i'] = \{e\}$.

Proof sketch. Since K_i is abelian, elements of $[\Gamma_i', \Gamma_i']$ are pure translations. Take

$$g = (A, b) \in \Gamma_i', \qquad h = (I, c) \in [\Gamma_i', \Gamma_i'].$$

Then $[g,h] = (I, (A-I)c)$. On the complement of its kernel, $A - I$ is invertible. If $(A-I)c \neq 0$, then $(A-I)^k c \neq 0$ for every k, and

$$[g,[g,\ldots,[g,h]\ldots]] = (I, (A-I)^k c) \neq e,$$

contradicting nilpotency. □

2.3 No rotation

Corollary 2.5. *G_0 is nilpotent.*

Proof sketch. For $g, h, k \in G_0$, take $g_i, h_i, k_i \in \Gamma_i'$ converging to g, h, k. Then $[[g_i, h_i], k_i] = e$ is ε_i-close to $[[g, h], k]$, so $[[g, h], k] = e$. □

Corollary 2.6. *$G_0 = \mathbb{R}^n$ consists of pure translations.*

Proof sketch. G_0 is a connected nilpotent Lie group, so every compact subgroup is in its center. The stabilizer

$$(G_0)_p = \{g \in G_0 : gp = p\}$$

is compact, hence $[(G_0)_p, G_0] = e$. Since G is transitive and $|G/G_0| < \infty$, G_0 is transitive. For $q \in \mathbb{R}^n$, choose $h \in G_0$ with $hp = q$. If $g \in (G_0)_p$, then

$$gq = ghp = hgp = hp = q.$$

Thus $g = e$, the action is free, and $G_0 = \mathbb{R}^n$ consists of pure translations. □

Corollary 2.7. *$\pi_{O(n)}(\Gamma_i')$ is trivial.*

Proof sketch. Suppose it is nontrivial. Put $c = |G/G_0|$ and take $\varepsilon \ll 1/c$. Choose a small element with nontrivial rotational part. The finite-index relation

$$|\Gamma_i/\Gamma_i'| = c$$

allows a power to lie in Γ_i'. Identifying the groups with $\pi_1(M, p)$ and passing to a subsequence gives a fixed element $g = (A, b)$ with nontrivial rotational part appearing in Γ_i'. Under the rescaling its limit is $\pi_{O(n)}(g)$, contradicting the fact that the limit lies in G_0, the pure translation group. □

Summary of the proof.

1. Take the equivariant limit $(\widetilde{M}_i, \tilde{p}_i, \Gamma_i) \xrightarrow{\text{eGH}} (\mathbb{R}^n, 0, G)$ and $\Gamma_i' \triangleleft \Gamma_i$ with $|\Gamma_i/\Gamma_i'| = |G/G_0| < \infty$.
2. The commutator estimate gives nilpotency of Γ_i'. Control its nilpotency steps to obtain nilpotency of G_0, and hence $G_0 = \mathbb{R}^n$.
3. The groups Γ_i' consist of pure translations.

3 Gromov's theorem

3.1 Set-up: pseudo covers and local groups

Consider $\exp_{p_i} : T_{p_i}M_i \to M_i$ with $|\operatorname{Sec}_{M_i}| \le \varepsilon_i$. The exponential map is locally isometric at v for $|v| < \pi/\sqrt{\varepsilon_i}$. Choose

$$R_i \to \infty, \qquad R_i \ll \frac{1}{\sqrt{\varepsilon_i}}, \qquad M_i' = B_{R_i}(0) \subset T_{p_i}M_i.$$

The map $\exp : M_i' \to M_i$ is called a pseudo cover.

Let $G_i(p_i, 2R_i)$ be the set of all geodesic loops at p_i with length at most $2R_i$. Such a loop corresponds to a ray from 0 to a point in

$$\exp^{-1}(p_i) \cap B_{2R_i}(0).$$

For $\alpha, \beta \in G_i(p_i, 2R_i)$, lift their concatenation to define the product $\alpha\beta$.

The elements of $G_i(p_i, 2R_i)$ give isometric maps on M_i'. For $\alpha \leftrightarrow g$ with lifted endpoint q, $g(0) = q$. Its differential is described by

$$T_0 M_i' \xrightarrow{d\exp} T_{p_i} M_i \xrightarrow{\text{parallel transport along } \alpha} T_{p_i} M_i \xrightarrow{(d\exp_q)^{-1}} T_q M_i'.$$

Extend $g : M_i' \to B_{3R_i}(0)$. We have

$$M_i'/G_i(p_i, 2R_i) = M_i,$$

and

$$\begin{array}{ccc} (M_i', 0, G_i(p_i, 2R_i)) & \xrightarrow{\text{eGH}} & (\mathbb{R}^n, 0, G) \\ \downarrow & & \downarrow \\ M_i & \xrightarrow{\text{GH}} & \{\text{pt}\}. \end{array}$$

There is a normal local subgroup $G_i' \triangleleft G_i(p_i, 2R_i)$ such that

$$G_i' \longrightarrow G_0, \qquad |G_i(p_i, 2R_i)/G_i'| = |G/G_0| < \infty.$$

The quotient M_i'/G_i' is a finite covering space of M_i. For every $\varepsilon > 0$ and all large i,

$$\langle G_i' \cap B_\varepsilon(e) \rangle = G_i'.$$

The group generated by the local group. Let $\widehat{G}_i$ be the abstract group generated by $G_i(p_i, 2R_i)$. There is a surjective map

$$\pi : \widehat{G}_i \longrightarrow \pi_1(M_i, p_i).$$

Use $\widehat{G}_i$ and M_i' to glue a space $\widehat{M}_i$. This gives

$$(\widehat{M}_i, \widehat{G}_i) \longrightarrow (\widetilde{M}_i, \pi_1(M_i, p_i)) \longrightarrow M_i,$$

and $\widehat{M}_i = \widetilde{M}_i$, $\widehat{G}_i = \pi_1(M_i, p_i)$. The map $G_i(p_i, 2R_i) \to \widehat{G}_i$ will be shown to be injective.

3.2 Commutator estimates

Holonomy. For $\alpha \leftrightarrow g \in G_i'$, write

$$m(\alpha) = (r(\alpha), t(\alpha)) \in O(n) \ltimes \mathbb{R}^n.$$

Here $r(\alpha)$ is parallel transport along α, and $t(\alpha) = \alpha'(1)$. Being close to e corresponds to $m(\alpha)$ being small.

Lemma 3.1.

$$\begin{aligned} d(r(\alpha)r(\beta), r(\alpha\beta)) &\le \varepsilon_i |t(\alpha)|\,|t(\beta)|, \\ |t(m(\alpha)m(\beta)) - t(\alpha\beta)| &\le \varepsilon_i |t(\alpha)|\,|t(\beta)|(|t(\alpha)| + |t(\beta)|). \end{aligned}$$

Proof sketch. Use the curvature integral over the geodesic triangle; its area is bounded by a constant times $|t(\alpha)||t(\beta)|$. □

In the notation of the notes,

$$d(m(g)m(h), m(gh)) \le \varepsilon_i \, d(e, m(g)) d(e, m(h)).$$

Lemma 3.2 (Commutator estimate)**.** *There is $C > 1$ such that, for $\varepsilon \ll 1/C$ and all sufficiently large i, if $\alpha, \beta \in G_i' \cap B_\varepsilon(e)$, then*

$$d(e, m([\alpha, \beta])) \leq C\, d(e, m(\alpha)) d(e, m(\beta)).$$

Proof sketch. Use the previously proved estimate

$$d(e, [m(\alpha), m(\beta)]) \leq C\, d(e, m(\alpha)) d(e, m(\beta))$$

and

$$d(m([\alpha, \beta]), [m(\alpha), m(\beta)]) \leq \varepsilon_i\, d(e, m(\alpha)) d(e, m(\beta)).$$

□

Corollary 3.3. *G_i' is nilpotent.*

3.3 Controlling the nilpotency steps

Short generators. Choose $g_{i1} \neq e$ so that $d(m(g_{i1}), e)$ is smallest. Then $[g_{i1}, G_i'] = e$. Choose $g_{i2} \in G_i' \setminus \langle g_{i1} \rangle$ so that $d(m(g_{i2}), e)$ is smallest. Then

$$[g_{i2}, G_i'] \subset \langle g_{i1} \rangle.$$

Continuing, obtain $g_{i1}, \ldots, g_{ik_i}$ with triangular commutator relations. The nilpotency step of G_i' is at most k_i.

Lemma 3.4. *$k_i \leq C(n)$.*

Proof sketch. If k_i is too large, there are $1 \leq j < k \leq k_i$ such that

$$\frac{\log m(g_{ij})}{|\log m(g_{ij})|} \quad \text{is close to} \quad \frac{\log m(g_{ik})}{|\log m(g_{ik})|}.$$

Then

$$d(m(g_{ik} g_{ij}^{-1}), e) < d(m(g_{ik}), e),$$

a contradiction. □

Corollary 3.5. *G_0 is nilpotent, and $G_0 = \mathbb{R}^n$ consists of pure translations.*

3.4 Small rotations

Lemma 3.6 (Escape property)**.** *There is $\delta > 0$ such that each nontrivial $g_i \in B_\delta(e) \cap G_i'$ has a power $g_i^k \notin B_\delta(e)$.*

Proof sketch. Suppose $\langle g_i \rangle \subset B_\delta(e)$. The group $\langle g_i \rangle$ converges to a subgroup $H \subset G_0 \cap B_\delta(e)$. Since G_0 is a Lie group, choose δ small enough that $G_0 \cap B_\delta(e)$ contains no nontrivial subgroup. Then $H = e$. □

Lemma 3.7. *For $g_i \in G_i'$,*

$$d(r(g_i), e) \leq \varepsilon_i |t(g_i)|.$$

Proof sketch. Let g_i be ε_i-close to $g \in G_0 = \mathbb{R}^n$. For an element outside the fixed small ball, use

$$d(r(g_i), e) \leq \varepsilon_i, \qquad |t(g_i)| \geq \delta - \varepsilon_i.$$

For a small element, take the smallest $k > 0$ such that $g_i^k \notin B_\delta(e)$ and apply the estimate to g_i^k. The power estimate divides the rotational size by k, while

$$|t(g_i)| \geq \frac{\delta - \varepsilon_i}{k}.$$

This gives the small-rotation estimate. □

3.5 Constructing a nilpotent basis

Fix a large i. Choose $g_1 \in G_i' \setminus \{e\}$ with $|t(g_1)|$ minimal. Choose $g_2 \notin \langle g_1 \rangle$ with the length of the projection of $t(g_2)$ onto $\{t(g_1)\}^\perp$ minimal. Continue by choosing $g_{k+1} \notin \langle g_1, \dots, g_k \rangle$ to minimize the length of the projection onto $\operatorname{span}\{t(g_1), \dots, t(g_k)\}^\perp$. Write

$$V_k = \operatorname{span}\{t(g_1), \dots, t(g_k)\}, \qquad H_k = \langle g_1, \dots, g_k \rangle.$$

Lemma 3.8. *If $P_{V_k^\perp} t(g) = 0$, then $g \in H_k$.*

Proof sketch. If $P_{V_{k-1}^\perp} t(g) \neq 0$, it is parallel to $P_{V_{k-1}^\perp} t(g_k)$. Choose an integer ℓ so that the projection of $t(g g_k^\ell)$ is shorter than the minimal projection for g_k. Then $g g_k^\ell \in H_{k-1}$, and hence $g \in H_k$. □

The set $\{t(g) : g \in G_i'\}$ is ε_i-dense. Thus there are exactly n generators $g_1, \dots, g_n$ such that

$$t(g_1), \dots, t(g_n) \text{ are linearly independent}, \qquad \langle g_1, \dots, g_n \rangle = G_i'.$$

Lemma 3.9. *There is $c > 0$ such that, for $g \notin H_k$, there exists $h \in H_k$ with*

$$|P_{V_k^\perp} t(gh)| \geq c\, |t(gh)|.$$

Proof sketch. For $k = 1$, choose ℓ so that

$$|P_{V_1} t(g g_1^\ell)| \leq \left(\frac{1}{2} + \varepsilon_i\right) |t(g_1)|.$$

Since $|t(g g_1^\ell)| \geq |t(g_1)|$, the perpendicular component controls the total length. For $k = 2$, first choose a power of g_2 in $\{t(g_1)\}^\perp$, then a power of g_1, obtaining

$$|P_{V_2} t(g g_2^s g_1^\ell)| \leq C\, |P_{V_2^\perp} t(g g_2^s g_1^\ell)|.$$

Continue in this way. □

Remark 3.10. The generators can be chosen so that

$$|t(g_k)| \leq c^{-1} |P_{V_{k-1}^\perp} t(g_k)|.$$

Theorem 3.11 (Nilpotent basis)**.** *For $1 \leq j < k \leq n$,*

$$[g_k, g_j] \in \langle g_1, \dots, g_{j-1} \rangle.$$

Proof sketch.

$$|t([g_k, g_j])| \ll |t(g_j)| \leq c^{-1} |P_{V_{j-1}^\perp} t(g_j)|.$$

The minimal choice of g_j gives the result. □

Any relation between $g_1, \dots, g_n$ can be written as

$$g_n^{a_n} g_{n-1}^{a_{n-1}} \cdots g_1^{a_1} = e.$$

The linear independence of $t(g_n), \dots, t(g_1)$ gives $a_j = 0$ for all j. Thus G_i' is a subset of a torsion-free nilpotent group Γ generated by n elements. The abstract group generated by G_i' is Γ, and $G_i' \to \Gamma$ is injective. Moreover,

$$\Gamma = \pi_1(M_i'/G_i'),$$

and the unit ball in M_i' is isometric to the corresponding unit ball in the universal cover.

3.6 Constructing a nilmanifold

Lemma 3.12 (Mal'cev)**.** Γ *embeds in an* n*-dimensional connected, simply connected nilpotent Lie group* N*.*

Metric. Take $v_1, \dots, v_n$ in the Lie algebra such that $\exp_N(v_j) = g_j$. Define the metric by

$$\langle v_j, v_k \rangle = t(g_j) \cdot t(g_k).$$

Local diffeomorphism. Use $\exp_N$ to construct a local diffeomorphism

$$f : B_1(p_i') \longrightarrow N.$$

For $g \in G_i'$, $gf(x)$ is ε_i-close to $f(gx)$. Use the center of mass to obtain a global diffeomorphism

$$f : \widehat{M}_i \longrightarrow N$$

that is Γ-equivariant. It descends to a diffeomorphism

$$\widehat{M}_i/\Gamma \cong N/\Gamma.$$

4 Ricci curvature

4.1 Set-up

In Theorem 3, consider

$$\operatorname{diam}(M_i) \le \varepsilon_i, \qquad \operatorname{Vol}(B_1(\tilde{p}_i)) \ge v > 0.$$

The conclusion is that a finite covering space is a nilmanifold. The equivariant set-up is

$$\begin{array}{ccc} (\widetilde{M}_i, \tilde{p}_i, \Gamma_i) & \xrightarrow{\text{eGH}} & (\mathbb{R}^n, 0, G) \\ \big\downarrow & & \big\downarrow \\ M_i & \xrightarrow{\text{GH}} & \{\text{pt}\}, \end{array}$$

with $\Gamma_i' \triangleleft \Gamma_i$ and $\Gamma_i/\Gamma_i' \cong G/G_0$.

4.2 Escape norm

For $g_i \in \Gamma_i' \cap B_\delta(e)$, define

$$|g_i| = \frac{1}{k}, \qquad k = \min\{j > 0 : g_i^j \notin B_\delta(e)\}.$$

Commutator estimate. There is $C > 0$ such that, for $\varepsilon \ll 1/C$,

$$|[g, h]| \le C|g|\,|h|.$$

Proof sketch. For an equivariant approximation $\varphi_i : \Gamma_i' \to G_0$, use

$$d(e, [\varphi_i(g), \varphi_i(h)]) \le C\, d(e, \varphi_i(g)) d(e, \varphi_i(h)).$$

If $|g|, |h| \gg \varepsilon_i$, compare g, h with $\varphi_i(g), \varphi_i(h)$. If $|g|, |h| \lesssim \varepsilon_i$, choose s, t so that g^s, h^t first leave $B_\delta(e)$. Redefine the maps using

$$\varphi_i(g) = \frac{1}{s}\varphi_i(g^s), \qquad \varphi_i(h) = \frac{1}{t}\varphi_i(h^t).$$

The comparison at the scale of g^s, h^t has error ε_i; at the scale of g, h, the indicated error is $\varepsilon_i/(st)$.